

Autonomous UAV Navigation and Mapping for Accurate Fruit Detection and Counting in Controlled Environments: Simulation and Real-World Validation

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Abstract—This paper presents a solution for high accuracy fruit counting in a Controlled Environment Agriculture (CEA) settings using Unmanned Aerial Vehicles (UAV) implemented in simulation and real-world scenarios. Using the fine-tuned YOLOv8 [1] model for precise object detection and classification along with a custom path planning algorithm for simulation. This approach integrates a hybrid A* Traveling Salesman Problem (TSP) algorithm for efficient 3D path planning. The solution is further extended to the real-life scenario using LiDAR based mapping and point-cloud filtering techniques to avoid recounting of fruits. The suggested methodology increases operating efficiency, reduces dependency on human labor, and improves accuracy. Experimental results, derived from both simulations and real-world testing, achieving a fruit-count accuracy of 98% and 90% respectively, demonstrate the effectiveness of this integrated approach. This solution was implemented in the International Conference of Unmanned Aircraft System (ICUAS) [2] UAV Competition 2024, securing 2nd position in the simulation phase (out of 24 teams) and 3rd overall in the real-world phase.

I. INTRODUCTION

Agriculture has evolved significantly over the centuries, transitioning from traditional, labor-intensive practices to highly autonomous and data-driven systems. Controlled Environment Agriculture (CEA) [3], which involves growing crops in optimized, enclosed settings, has emerged as a promising approach to meet global food demands sustainably. The development in the sector of Unmanned Aerial Systems in recent years has shown promising progress within agricultural domains [4].

This paper introduces an approach that utilizes Unmanned Aerial Vehicles (UAV) to accurately maneuver, detect, localize and count fruits in a highly controlled environment such as a greenhouse.

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The problem statement is defined as a task, for a UAV to autonomously count different kinds of fruit in specific plant beds in a CEA. Focusing on high accuracy, minimizing time, and total path to be taken.

Our contributions include: (1) A custom hybrid A* Traveling Salesman Problem (TSP) algorithm [5] for computationally efficient path planning in structured CEA settings reducing path length and travel time. (2) LiDAR-Camera fusion to globally localize the region of interest (here fruit) along with tracking fruits over a series of frames to improve fruit-count accuracy (3) Pipeline which achieves 98% fruit-count accuracy in simulation and 90% accuracy in real-world greenhouse scenario.

We initially ground our proposed methodology by implementing it in a simulated environment. We interpret the environment as a graph-based problem by considering individual plant beds as graph nodes with connections based on their relative spatial positions. On successful creation of nodes, it can be viewed as a single-agent multiple-objective problem, analogous to the TSP. Informed search methods are used that are computationally optimized by incorporating a heuristic, like time, to discover the most efficient path to achieve the goal.

Once the UAV reaches the designated location. Data collected from IMU, RGB and Depth Camera undergoes image angle correction and depth segmentation [6], after which it is sent to a fine-tuned YOLOv8 model [1] for object detection. Subsequently, the detected objects are processed by a custom algorithm that accurately localizes and counts them.

The work was validated through the 2024 International Conference on Unmanned Aircraft Systems' UAV Competition [2], where our team secured 2nd position among 24 international teams [7]. The simulated setting is shown in Figure 1. Accurate fruit counting, amount of time spent traversing, and path length were used as evaluation criteria for the competition.

The challenge environment runs on a Linux Ubuntu 20.04 LTS server running ROS Noetic (Robot Operating System) in Gazebo11.

The second phase of our methodology focuses on implementation in real-world scenario, in a non-ideal environment. For such environments, this approach includes visual tracking of fruit and using LiDAR Camera fusion to localize fruits in global frame ensuring consistent 3D world coordinates independent of the orientation of the UAV. This approach

ensures that the same fruit is not counted multiple times. Further point cloud filtering techniques were implemented to improve accuracy.

This approach was extensively tested using data from the 2024 ICUAS UAV Competition final phase. The data comprised an RGB camera feed and its corresponding LiDAR, IMU and GPS data in the form of ROSBAG files. The data was collected from UAV operating in a real-world agricultural environment featuring fruits in yellow and red.

II. LITERATURE REVIEW

Accurate fruit detection and counting in controlled agricultural environments, such as greenhouses, has been an active area of research due to its potential to automate labor intensive tasks and optimize yield estimation.

Traditional computer vision techniques, such as those by [8], relied on fruit detection and color segmentation. However, these approaches tend to fail under partial occlusions and varying lighting conditions. To address these limitations, a combination of YOLOv8-s model and multi-frame fruit tracking is utilized, showing reliable results in both simulation and real-world settings.

For UAV path planning in 3D environments, When determining the optimal sequence of nodes to visit, the problem can be formulated as a TSP. Existing solutions such as brute-force search and genetic algorithms tend to be computationally intensive and greedy algorithms do not provide satisfactory approximations. To overcome these issues, a hybrid approach combining A* [9], the Minimum Spanning Tree algorithm [10], and the Christofides algorithm [11] is proposed, which yields near-optimal solutions with significantly reduced computational complexity.

In context of 3D mapping, researchers have widely used LiDAR-Camera sensor fusion approach [12] instead of monocular depth estimation [13] for high precision fruit localization and mapping, but are prone to drift and noise, which this paper solves by assigning confidence values to each fruit cluster tracked over a series of frames, ensuring high precision.

III. METHODOLOGY

We build our solution in two stages: The first stage models the problem statement in a simulation environment, this facilitates iterative evaluation of the path planning algorithms and basic testing for fruit detection. The second stage explores transfer of our solution to real-world scenarios.

A. Simulation

The orchard environment, as shown in Fig. 1 features 27 plant beds arranged in 3 rows, and stacked across 3 levels. Each plant bed contains up to three plants with three fruit varieties: Tomato (red), Pepper (yellow), and Eggplant (purple). The front and back side of the plants are asymmetrical, making it possible for a fruit to be visible from the front but not from the back, or from both sides.

The UAV is permitted to navigate along and around the shelves but is restricted from flying between or directly above the plant beds.

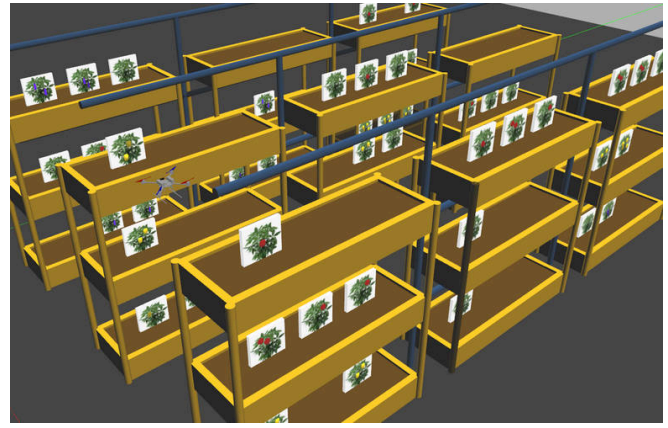


Fig. 1. Simulation Environment

An orchard farm is a systematic and static environment, which allows the use of exact layout and coordinate system of the orchard environment in simulation as well as real-world scenarios. This approach involves simplifying the environment into a grid structure, where each node represents the location of points of interest. The point of interest can be the front or back of the plant and the side nodes to go to the next row of plant beds. The edges connecting these nodes represent permissible paths for the UAV to travel.

There are weighted nodes for each plant bed, front and back separately ($2 \times n$ nodes for n plant beds), and 6 nodes each on the left and right ends of each row, for traversing to the next row ($12 \times r$ nodes for r rows). For this scenario consisting of 27 beds and 3 rows, there are a total of 90 nodes. These nodes are calculated as the environment is known a priori and are plotted as shown in Fig. 2.

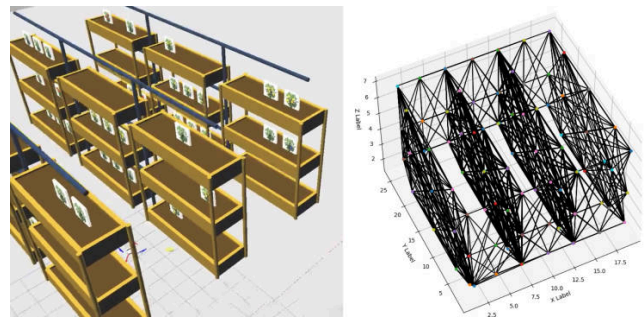


Fig. 2. Environment simplified to grid structure

1) *Path Planning*: It is essential to count the maximum number of fruits in the least amount of time.

Factors considered in optimality include: 1) Temporal Efficiency : Reducing the time required by UAV to reach nodes. 2) Energy Consumption: Reducing path length and ensuring minimum altitude changes. 3) Computation Resources: Optimizing for efficient use

The aim is to find the best route to visit the front and back of n different plant beds. Path planning is reduced to two sub problems which include finding the shortest path between any 2 nodes along the permissible edges, and the TSP that is to formulate the sequence of nodes to be visited

for the optimal path.

a) *Determining the optimal path:* To find the most optimal path between any 2 nodes, we use the A* algorithm [14] with weighted nodes and multiple heuristic functions.

The algorithm consists of 3 key heuristic functions: The combined heuristic $h(n, g)$ is given by:

$$h(n, g) = h_d(n, g) + h_{\text{yaw}}(n) + h_{\text{altitude}}(n) \quad (1)$$

- 1) **Euclidean Distance (h_d):** Euclidean distance between the current node and goal

$$h_d(n, g) = \sqrt{(x_n - x_g)^2 + (y_n - y_g)^2 + (z_n - z_g)^2} \quad (2)$$

- 2) **Yaw Requirement (h_{yaw}):** Cost associated with necessary yaw adjustments, k_{yaw} denotes the UAV's yaw angle, expressed in radians.

$$h_{\text{yaw}}(n) = k_{\text{yaw}} \cdot |((\text{yaw}_{n+1} - \text{yaw}_n + \pi) \bmod 2\pi) - \pi| \quad (3)$$

- 3) **Altitude Weight (h_{altitude}):** Higher weights are assigned to nodes at greater altitudes. Here, z_n denotes the altitude of the UAV at the n^{th} node.

$$h_{\text{altitude}}(n) = k_{\text{altitude}} \cdot (z_{n+1} - z_n) \quad (4)$$

The three heuristics are designed with real-world scenarios in mind. Gaining altitude is more time and energy-consuming than descending. In addition, yaw plays a crucial role in time efficiency. Since the UAV must assess both the front and back of the orchard, it occasionally needs to perform a full 180-degree yaw. By factoring in yaw beforehand, we can command the UAV to yaw during its approach to the desired point, optimizing its path planning accordingly, and by considering it as a heuristic, the proposed algorithm can take better consideration when to prefer a node with yaw for an optimal solution.

b) *Determining the sequence of nodes:* For scenarios requiring inspection of specific plant beds (e.g. fruits at different growth stages), it is essential to determine the optimal sequence for visiting the nodes, minimizing both the total distance traveled and the time taken.

An informed approach is used since traversing the farm row by row is the most efficient method, as completing one row before moving to the next minimizes unnecessary travel. Movement between rows can only occur by exiting at the end of a row, as transitions from the middle are not possible. Returning to a previous row would inevitably increase travel time, so this sequential approach optimizes overall efficiency.

This approach enables a structured traversal of nodes by dividing the farm into distinct segments, traversing each horizontal corridor individually. The algorithm is applied separately to each face.

First we generate a distance matrix for the desired points, where the values in the matrix represent the total weight of the traveled path proposed by the modified A* algorithm. Further we implement a comprehensive optimization informed approach using a combination of Minimum Spanning Tree (MST) and Christofides algorithm.

On nodes of each face, by applying a Minimum Spanning Tree, we determine the initial sequence of nodes to visit. However, this resulted in some nodes having multiple paths from the same node. To address this issue, Christofides algorithm ensures that each node has only two connections, allowing the UAV to maintain a continuous path. This approach leads to highly optimal path due to the simplicity of the environment.

This process returns the sequence of nodes that the UAV should visit on each face and A* Algorithm returns the path between respective nodes. Combining this, a mission is created using the resulting list of nodes and their corresponding yaw angles to be sent to the UAV.

This algorithm reduces the computational complexity significantly for 3D path planning while maintaining robust performance in structured static orchard environments.

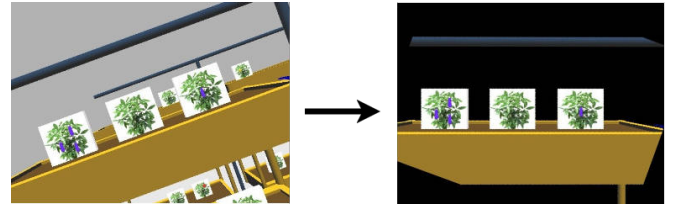


Fig. 3. Application of Depth based filtering and IMU correction on the data captured in the environment.



Fig. 4. Localizing fruits on multiple views of the same plant bed in the simulation environment.

2) *Perception:* Once arrived at the designated plant bed, the task is to accurately count the number of fruits. The simulation involves three classes of fruits, each represented by a different color. (Tomato (red), Pepper (yellow), and Eggplant (purple)). These fruit beds need to be assessed from two perspectives: front and back. An additional task is to localize the fruits. This step is essential for identifying fruits that are visible from both sides, which may otherwise lead to an inaccurate count for each fruit class.

Since the simulation provides ideal conditions with noiseless data, the model performs accurately with minimal false positives. However, the absence of noise also results in all fruits within the frame being detected, even those located far from the area of interest.

Depth-based filtering is employed using data from a depth camera. Accurate threshold values were determined based on the distance between successive fruit shelves and the variance in UAV positioning, allowing precise segmentation of objects of interest from the rest of the frame, illustrated in Fig. 3.

For orientation correction, the UAV's Inertial Measurement Unit (IMU) data is used to obtain the orientation at the

TABLE I

EVALUATION OF OUR APPROACH IN DIFFERENT SCENARIOS WITHIN THE GAZEBO ENVIRONMENT. EACH PLANT BED IS ASSIGNED A NUMBER, AND THE BEDS COLUMN INDICATES THE ONES TO BE VISITED

Sim	Beds	GT Fruits	Predicted	Accuracy
1	8,9,11,16,21,25	12	12	100
2	8,9,16	0	0	100
3	1,15,16,20,21,27	12	12	100
4	3,10,12,23,25,27	11	11	100
5	1,6,13,15,17,19,25	5	5	100
6	1-27	35	33	94.28
7	2,3,4,16,20,24,26	18	18	100
8	1,2,6,7,13,14	22	21	95.45
9	2,3,17,18,20,21,23	17	17	100
10	1,3,5,7,8,13,25	15	15	100
11	4,12,14,17,18,19,22	11	11	100
12	1,3,10,21,22,24,25	15	13	86.67
Mean Accuracy (%):				98.18

time of image capture, which helps rotate the image so that plants are properly aligned within the frame. After applying depth segmentation and orientation correction, the YOLO model provides effective localization and classification of the plant beds and fruits.

As the drone reaches a target location, multiple images are captured for pre-processing and detection. The images are filtered based on the number of plant beds present, removing any outlier images where the plant bed count deviates from the median. A similar process is applied to filter images based on fruit counts.

The UAV captures both front and back views of each plant during its visit. These are matched using known ground truth plant positions, and a final image is selected for each view after processing. To determine the effective fruit count, the distance from the front fruit to the left and upper edges of the plant bed, and from the back fruit to the right and upper edges, is considered. If the difference is within a specific threshold, the fruit is counted as the same across both views. To further refine this method and avoid double-counting, the difference in the ratio of distance to width/height is used instead, effectively locating fruits relative to the plant bed. (as shown in Fig. 4)

Now, the fruit count of a shelf is stored, and the drone moves to next shelf. The fruit count of all the shelves is added to give the final fruit count.

3) *Results*: The Gazebo-based simulation environment provided an ideal space for the development of the path planning algorithm which can be directly applied to any Controlled Environment Agriculture setting resembling the simulation environment. We were able to test both the Path planning and the perception algorithm multiple times in different cases and scenarios (as shown in Table I). The Perception algorithm pre-processes the data so that the YOLOv8-S model can run effortlessly. The fruit counting algorithm prevents any double counting of the fruits, giving an accuracy of 98.18%

The Results are showcased in Fig. 5. A cumulative score

Rank	Team	Eval1	Eval2	Eval3	Eval4	Eval5	Eval6	Bo4 AVG
1	ITUAV	0	89.76	97.33	98.73	0	94.65	95.12
2	UAS-DTU	0	0	97.91	92.21	96.62	92.45	94.80
3	ASL	0	84.89	94.65	85.11	97.18	91.71	92.16
4	AIROLab	0	0	94.25	93.45	77.28	86.7	87.92
5	AVANT-UFMG	0	0	86.44	83.68	88.58	84.5	85.80
6	AVADERAGH	0	32.03	90.49	89.9	72.13	86.8	84.83
7	UMIC	0	57.70	73.86	93.71	85.68	38.16	77.74
8	SDCNLab	0	72.06	79.96	72.26	69.04	70.25	73.64
9	CVAR	63.05	64.43	70.58	79.7	66.84	66.46	70.40
10	FeikAir	0	0	75.54	68.33	71.05	65.82	70.19

Fig. 5. Performance evaluation of this approach on unseen configurations in the ICUAS'24 UAV Competition.

of 94.80 was achieved, which combines the following factors - Fruit Count Accuracy, Total Time Taken and Total distance traversed.

B. Real-World Scenario

This phase of the methodology focuses on real-world implementation. In contrast to the idealized simulation environment, it introduces the challenge of noisy data leading to double counting of fruits and inaccurate detection. To address these challenges, we propose solutions ensuring system can perform effectively under real-world conditions.

To avoid double counting of the same fruit, we localize the fruits in global coordinate frame. As a result, if the drone revisits the same fruit, the system recognizes it as already detected, preventing any redundant counting.

1) *Hardware Configuration*: We use a Kopterworx Hawk quadcopter equipped with an integrated Intel RealSense D415 camera [15] for capturing visual data and a Velodyne VLP-16 LiDAR [16] for depth sensing. An RTK-enabled GPS module provides centimeter-level positioning accuracy. The data is recorded from the IMU, GPS, LiDAR, and the video feed from the camera in a ROSBAG, along with their timestamps to ensure synchronization.

2) *Coordinate Transformation*: In real-world scenarios, the GPS provides latitude and longitude of the drone but for easier analysis, we convert it to local ENU (East North Up) Frame [17] in meters with respect to the origin as the UAV takeoff position geo coordinates. We further transform the ENU coordinates by incorporating the known offset between the GPS module and the LiDAR to ensure that coordinates are in LiDAR's reference frame.

3) *Mapping*: To accurately map LiDAR points in the ENU Coordinate frame. The LiDAR 3D point cloud captured in UAV's frame is rotated based on UAV's roll, pitch and yaw angles obtained from onboard IMU.

The rotation matrices [18] are applied to point cloud to account for the UAV's orientation in the local ENU frame. The full rotation is represented as:

$$\mathbf{P}_{\text{rotated}} = R_{\text{yaw}} \cdot R_{\text{pitch}} \cdot R_{\text{roll}} \cdot \mathbf{P} \quad (5)$$

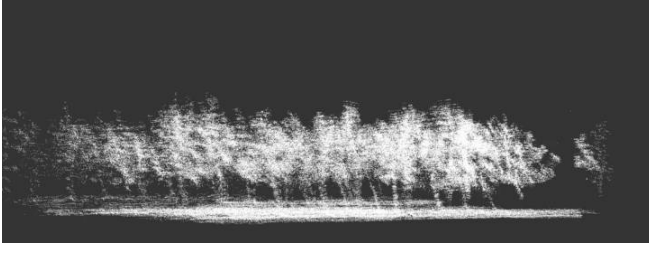


Fig. 6. Environment mapping demonstrated by registering the transformed point cloud over time

Where R_{roll} , R_{pitch} , R_{yaw} are the rotation matrices corresponding to the roll, pitch, and yaw angles of the UAV, and $\mathbf{P}_{\text{rotated}}$ is the transformed point in the rotated frame. The transformed point cloud coordinates are added to the corresponding UAV coordinates obtained from section III-B.2. After plotting the transformed point cloud over time, we get a 3D map of the environment (as shown in Fig. 6)

4) *Camera-LiDAR Fusion*: 3D coordinates with respect to each pixel of camera image are calculated using Camera-LiDAR Fusion. The first step is calibration [19]. It establishes a common reference frame between the sensors. Calibration enables combining point cloud data from LiDAR and visual frames from the camera feed. Intrinsic calibration was performed using an AprilTag grid for estimation of these parameters. $\mathbf{K}_C$ represents the camera's intrinsic matrix. For extrinsic calibration, the transformation between LiDAR and camera coordinates is represented by:

$$Z_C \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}_C [\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} X_L \\ Y_L \\ Z_L \\ 1 \end{bmatrix}. \quad (6)$$

where $\mathbf{R}$ and $\mathbf{t}$ denote the rotation matrix and translation vector between LiDAR and camera coordinates, respectively.

In the above equation, X_L , Y_L , and Z_L represent x , y , and z corresponding to the LiDAR's frame of reference. Using this approach, we achieved accurate Camera-LiDAR calibration as shown in Fig. 7.



Fig. 7. Demonstration of fruit detection in image frame, followed by localization using coordinate information derived from Camera-LiDAR fusion.



Fig. 8. 3D map showcasing detected fruits within a controlled orchard environment, with each cluster of points representing an individual fruit.

5) *Perception*: The Real-World implementation required the detection of two kinds of fruit - Yellow and Red. For detection and classification, the pre-trained YOLOv8-S model was used and fine-tuned on data collected from the greenhouse. The model offers a minimal accuracy-speed tradeoff compared to larger variants. It is able to detect both fruits with high accuracy, as shown in Fig. 7; however, some background regions were misclassified as yellow, leading to a few false positives. The confidence scores of such detections were low, thus a confidence threshold value was chosen to filter out these redundant detections.

6) *Fruit Tracking and Filtering*: The pixels corresponding to detected fruits are used to derive their 3D coordinates, as explained in section III-B.4. These coordinates are then mapped to form 3D fruit clusters in ENU frame, following the methodology detailed in section III-B.3. Further clustering techniques are applied to count the number of fruits. However, due to the dense and bushy nature of the plants, even slight delays in data synchronization lead to false mapping of branches or leaves in place of fruits, leading to an increase in False Positives.

To solve this issue, the center pixel value of each fruit in a camera frame was tracked for 30 consecutive image frames. The pixel value and its corresponding 3D coordinate were saved in memory temporarily. If the change in center pixel value was less than a certain threshold, it made it certain that all the 3D points collected in those respective frames correspond to the same fruit. If the number of points exceeded a threshold of 15 points for each tracked fruit, they were assigned a fruit number, else they were discarded from memory.

The fruits that are assigned a number are now statistically filtered. Points deviating significantly from the mean were removed. This ensured that all faulty 3D points of branches or leaves are filtered out. Now, every new cluster is compared to an existing one, if the new cluster is overlapping another formed cluster, then it is certain that they are the same fruit, if not then the cluster is assigned a new fruit number. This ensured that a fruit is not counted twice in cases where tracking of fruit is lost or the UAV returns back to the same fruit later some point in time, or views the fruit from the back or side angle. This approach filtered noisy data to a great extent as seen in Fig. 8. The small yellow circles depict the noisy point cloud, and large yellow circles correspond to final fruit filtered from the noisy data.

7) *Results*: The YOLOv8 model produces precise bounding boxes of both kinds of fruits even in the case of

occlusions, which with the help of LiDAR-camera Fusion and localization gives the coordinates of the fruit. We were successfully able to count fruits in a real-life environment with a precision of 90.12% for red fruits and 87.21% for yellow fruits (shown in Table II) using the proposed approach by filtering out noisy data. A major challenge was fruits being partially occluded by leaves which led to the mapping of leaves in place of the fruit. This was solved by visually tracking the fruit for consecutive frames and filtering out points away from the dense fruit cluster. Our solution was also validated under non-ideal conditions in the ICUAS 2024 UAV Competition final phase achieving 3rd Position Overall.

TABLE II
EVALUATES OUR APPROACH FOR BOTH YELLOW AND RED FRUITS
ACROSS VARIOUS FRUIT CONFIGURATIONS

	Ground Truth		Detected		Precision (%)	
	Yellow	Red	Yellow	Red	Yellow	Red
Flight 1	12	8	14	10	85.71	80.00
Flight 2	14	14	16	13	87.50	92.86
Flight 3	15	11	17	12	88.24	91.67
Flight 4	11	13	13	12	84.62	92.31
Flight 5	18	16	20	15	90.00	93.75
Mean					87.21	90.12

IV. CONCLUSION

In conclusion, we present our solution that effectively combines multiple approaches to provide a complete UAV system for automating agriculture activities. Our base methodology was deployed and tested in a simulated environment, helping us to identify the most efficient path-planning strategies. The initial implementation in the simulation also allowed us to refine our basic formulation of the environment. For the task of fruit counting and localization the simulation provided an ideal environment and therefore we observe almost 100 % accuracy in each of the runs for the task of fruit counting and localization. This is due to the fact that all captured photos in the train and test times belong to the same domain causing almost no variation in data.

For the real-world scenario, the part of energy efficiency in path planning was directly transferrable. On the other hand, the challenging tasks include accurate camera-LiDAR calibration despite of slight changes in extrinsic properties. Another important aspect came in with correctly re-identifying the same fruits. To tackle the above mentioned, we presented point cloud filtering strategies and approaches that rely on independent visual and point cloud based tracking for accurate fruit counting.

The future aspects involve implementing the solution in different agricultural settings and making this algorithm more robust and accurate. We will also incorporate the ability to detect the condition of the fruit, its stages of growth, and the presence of insects and weeds to provide the farmer with the entire overview of the agricultural field.

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